

Research Monographaskdfjhjk

Book Author

Month Year

Mục lục

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Preface

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Phần I

First half of the book

Chương 1

Introduction

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1.1 Research question

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1.2 Answer to the question

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¹Excepteur sint, sunt in culpa qui officia deserunt mollit anim id est laborum. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.² Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

1.3 Positioning in the literature

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1.4 A section

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1.4.1 A subsection

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1.4.2 Yet another subsection

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1.5 A final introduction section

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Chương 2

Sections and content

Chương 3

Math

Phần II

Second half of the book

Chương 4

Cross-references and quotes

Chương 5

Thiết kế vật lý

5.1 Phân hệ Cấu trúc mã & Phân tích cây cú pháp trừu tượng (AST)

5.1.1 Cú pháp Rút gọn (Logical Schema Syntax)

- `commits`
(`commit_id` UUID PK,
`repository_id` UUID FK,
`branch_id` UUID FK,
`pr_id` UUID FK,
`commit_hash` VARCHAR(255),
`author_name` VARCHAR(255),
`author_email` VARCHAR(255),
`message` TEXT,
`committed_at` TIMESTAMP,
`created_at` TIMESTAMP,
`updated_at` TIMESTAMP)
- `pull_requests` (`pr_id` UUID PK, `repository_id` UUID FK,
`pr_number` INTEGER, `title` VARCHAR(500), `description` TEXT,
`source_branch` VARCHAR(255), `target_branch` VARCHAR(255),
`status` VARCHAR(20), `created_at` TIMESTAMP, `updated_at` TIMESTAMP, `merged_at` TIMESTAMP)
- `pull_request_labels` (`pull_request_label_id` UUID PK, `pr_id`

UUID FK, label_name VARCHAR(100), label_color VARCHAR(20))

- files (file_id UUID PK, repository_id UUID FK, file_path VARCHAR(1000), file_type VARCHAR(50), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP)
- code_diffs (diff_id UUID PK, commit_id UUID FK, pr_id UUID FK, file_id UUID FK, file_path VARCHAR(1000), change_type VARCHAR(20), additions INTEGER, deletions INTEGER, diff_content TEXT, created_at TIMESTAMP)
- code_entities (code_entity_id UUID PK, file_id UUID FK, entity_name VARCHAR(255), entity_type VARCHAR(50), signature TEXT, doc_comment TEXT, start_line INTEGER, end_line INTEGER, created_at TIMESTAMP, updated_at TIMESTAMP)
- dependencies (dependency_id UUID PK, source_entity_id UUID FK, target_entity_id UUID FK, dependency_type VARCHAR(50), created_at TIMESTAMP)

5.1.2 Mã DDL SQL Hoàn Chỉnh (schema_member_3.sql)

```
1
2 CREATE TABLE commits (
3     commit_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
4     repository_id UNIQUEIDENTIFIER NOT NULL,
5     branch_id UNIQUEIDENTIFIER NOT NULL,
6     pr_id UNIQUEIDENTIFIER NULL,
7     commit_hash VARCHAR(255) NOT NULL,
8     author_name VARCHAR(255) NULL,
9     author_email VARCHAR(255) NULL,
10    message NVARCHAR(MAX) NULL,
11    committed_at DATETIME2 NOT NULL,
12    created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
13    updated_at DATETIME2 NULL,
14
15    CONSTRAINT uq_commits_repository_hash
16        UNIQUE (repository_id, commit_hash)
17 );
18
19 CREATE INDEX idx_commits_repository_id
20     ON commits(repository_id);
21
22 CREATE INDEX idx_commits_branch_id
23     ON commits(branch_id);
24
```

```

25 CREATE INDEX idx_commits_pr_id
26     ON commits(pr_id);
27
28 CREATE INDEX idx_commits_committed_at
29     ON commits(committed_at);
30
31 CREATE TABLE pull_requests (
32     pr_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
33     repository_id UNIQUEIDENTIFIER NOT NULL,
34     pr_number INT NOT NULL,
35     title VARCHAR(500) NOT NULL,
36     description NVARCHAR(MAX) NULL,
37     source_branch VARCHAR(255) NULL,
38     target_branch VARCHAR(255) NULL,
39     status VARCHAR(20) NOT NULL DEFAULT 'OPEN',
40     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
41     updated_at DATETIME2 NULL,
42     merged_at DATETIME2 NULL,
43
44     CONSTRAINT uq_pull_requests_repository_pr
45         UNIQUE (repository_id, pr_number),
46
47     CONSTRAINT chk_pull_requests_status
48         CHECK (status IN ('OPEN', 'CLOSED', 'MERGED'))
49 );
50
51 CREATE INDEX idx_pull_requests_repository_id
52     ON pull_requests(repository_id);
53
54 CREATE INDEX idx_pull_requests_status
55     ON pull_requests(status);
56
57 CREATE INDEX idx_pull_requests_created_at
58     ON pull_requests(created_at);
59
60 CREATE TABLE pull_request_labels (
61     pull_request_label_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT
        NEWID(),
62     pr_id UNIQUEIDENTIFIER NOT NULL,
63     label_name VARCHAR(100) NOT NULL,
64     label_color VARCHAR(20) NULL,
65
66     CONSTRAINT uq_pull_request_labels
67         UNIQUE (pr_id, label_name)
68 );
69
70 CREATE INDEX idx_pull_request_labels_pr_id
71     ON pull_request_labels(pr_id);
72

```

```

73 CREATE INDEX idx_pull_request_labels_name
74     ON pull_request_labels(label_name);
75
76 CREATE TABLE files (
77     file_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
78     repository_id UNIQUEIDENTIFIER NOT NULL,
79     file_path VARCHAR(1000) NOT NULL,
80     file_type VARCHAR(50) NULL,
81     status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',
82     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
83     updated_at DATETIME2 NULL,
84
85     CONSTRAINT uq_files_repository_file_path
86         UNIQUE (repository_id, file_path),
87
88     CONSTRAINT chk_files_status
89         CHECK (status IN ('ACTIVE', 'DELETED', 'RENAMED'))
90 );
91
92 CREATE INDEX idx_files_repository_id
93     ON files(repository_id);
94
95 CREATE INDEX idx_files_file_path
96     ON files(file_path);
97
98 CREATE INDEX idx_files_file_type
99     ON files(file_type);
100
101 CREATE INDEX idx_files_status
102     ON files(status);
103
104 CREATE TABLE code_diffs (
105     diff_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
106     commit_id UNIQUEIDENTIFIER NOT NULL,
107     pr_id UNIQUEIDENTIFIER NULL,
108     file_id UNIQUEIDENTIFIER NOT NULL,
109     file_path VARCHAR(1000) NULL,
110     change_type VARCHAR(20) NOT NULL,
111     additions INT NULL,
112     deletions INT NULL,
113     diff_content NVARCHAR(MAX) NULL,
114     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
115
116     CONSTRAINT chk_code_diffs_change_type
117         CHECK (change_type IN (
118             'ADDED',
119             'MODIFIED',
120             'DELETED',
121             'RENAMED'

```

```

122        )),
123
124     CONSTRAINT uq_code_diffs_commit_file
125         UNIQUE (commit_id, file_id)
126 );
127
128 CREATE INDEX idx_code_diffs_commit_id
129     ON code_diffs(commit_id);
130
131 CREATE INDEX idx_code_diffs_pr_id
132     ON code_diffs(pr_id);
133
134 CREATE INDEX idx_code_diffs_file_id
135     ON code_diffs(file_id);
136
137 CREATE INDEX idx_code_diffs_change_type
138     ON code_diffs(change_type);
139
140 CREATE TABLE code_entities (
141     code_entity_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT
142     NEWID(),
143     file_id UNIQUEIDENTIFIER NOT NULL,
144     entity_name VARCHAR(255) NOT NULL,
145     entity_type VARCHAR(50) NOT NULL,
146     signature NVARCHAR(MAX) NULL,
147     doc_comment NVARCHAR(MAX) NULL,
148     start_line INT NULL,
149     end_line INT NULL,
150     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
151     updated_at DATETIME2 NULL,
152
153     CONSTRAINT chk_code_entities_entity_type
154         CHECK (entity_type IN (
155             'CLASS',
156             'INTERFACE',
157             'FUNCTION',
158             'METHOD',
159             'VARIABLE',
160             'CONSTANT',
161             'MODULE',
162             'PACKAGE',
163             'ENUM',
164             'STRUCT'
165         )),
166
167     CONSTRAINT chk_code_entities_line_range
168         CHECK (
169             start_line IS NULL
170             OR end_line IS NULL

```

```

170         OR start_line <= end_line
171     ),
172
173     CONSTRAINT uq_code_entities
174         UNIQUE (file_id, entity_name, entity_type, start_line)
175 );
176
177 CREATE INDEX idx_code_entities_file_id
178     ON code_entities(file_id);
179
180 CREATE INDEX idx_code_entities_entity_name
181     ON code_entities(entity_name);
182
183 CREATE INDEX idx_code_entities_entity_type
184     ON code_entities(entity_type);
185
186 CREATE TABLE dependencies (
187     dependency_id UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
188     source_entity_id UNIQUEIDENTIFIER NOT NULL,
189     target_entity_id UNIQUEIDENTIFIER NOT NULL,
190     dependency_type VARCHAR(50) NOT NULL,
191     created_at DATETIME2 NOT NULL DEFAULT GETDATE(),
192
193     CONSTRAINT uq_dependencies
194         UNIQUE (
195             source_entity_id,
196             target_entity_id,
197             dependency_type
198         ),
199
200     CONSTRAINT chk_dependencies_dependency_type
201         CHECK (dependency_type IN (
202             'CALLS',
203             'IMPORTS',
204             'INHERITS',
205             'IMPLEMENTS',
206             'USES',
207             'REFERENCES',
208             'DEPENDS_ON'
209         )),
210
211     CONSTRAINT chk_dependencies_no_self_reference
212         CHECK (source_entity_id <> target_entity_id)
213 );
214
215 CREATE INDEX idx_dependencies_source_entity_id
216     ON dependencies(source_entity_id);
217
218 CREATE INDEX idx_dependencies_target_entity_id

```

```

219         ON dependencies(target_entity_id);
220
221 CREATE INDEX idx_dependencies_dependency_type
222     ON dependencies(dependency_type);
223
224 GO
225
226 ALTER TABLE commits ADD CONSTRAINT fk_commits_repository
227     FOREIGN KEY (repository_id)
228     REFERENCES repositories(repository_id);
229
230 ALTER TABLE commits ADD CONSTRAINT fk_commits_branch
231     FOREIGN KEY (branch_id)
232     REFERENCES branches(branch_id);
233
234 ALTER TABLE commits ADD CONSTRAINT fk_commits_pull_request
235     FOREIGN KEY (pr_id)
236     REFERENCES pull_requests(pr_id)
237     ON DELETE SET NULL;
238
239
240 -- PULL REQUESTS
241
242 ALTER TABLE pull_requests ADD CONSTRAINT
243     fk_pull_requests_repository
244     FOREIGN KEY (repository_id)
245     REFERENCES repositories(repository_id);
246
247 -- PULL REQUEST LABELS
248
249 ALTER TABLE pull_request_labels ADD CONSTRAINT
250     fk_pull_request_labels_pr
251     FOREIGN KEY (pr_id)
252     REFERENCES pull_requests(pr_id)
253     ON DELETE CASCADE;
254
255 -- FILES
256
257 ALTER TABLE files ADD CONSTRAINT fk_files_repository
258     FOREIGN KEY (repository_id)
259     REFERENCES repositories(repository_id)
260     ON DELETE CASCADE;
261
262
263 -- CODE DIFFS
264
265 ALTER TABLE code_diffs ADD CONSTRAINT fk_code_diffs_commit

```

```

266     FOREIGN KEY (commit_id)
267     REFERENCES commits(commit_id)
268     ON DELETE CASCADE;
269
270 ALTER TABLE code_diffs ADD CONSTRAINT
271     fk_code_diffs_pull_request
272     FOREIGN KEY (pr_id)
273     REFERENCES pull_requests(pr_id)
274     ON DELETE SET NULL;
275
276 ALTER TABLE code_diffs ADD CONSTRAINT fk_code_diffs_file
277     FOREIGN KEY (file_id)
278     REFERENCES files(file_id)
279     ON DELETE CASCADE;
280
281 -- CODE ENTITIES
282
283 ALTER TABLE code_entities ADD CONSTRAINT fk_code_entities_file
284     FOREIGN KEY (file_id)
285     REFERENCES files(file_id)
286     ON DELETE CASCADE;
287
288 -- DEPENDENCIES
289
290 ALTER TABLE dependencies ADD CONSTRAINT
291     fk_dependencies_source_entity
292     FOREIGN KEY (source_entity_id)
293     REFERENCES code_entities(code_entity_id)
294     ON DELETE CASCADE;
295
296 ALTER TABLE dependencies ADD CONSTRAINT
297     fk_dependencies_target_entity
298     FOREIGN KEY (target_entity_id)
299     REFERENCES code_entities(code_entity_id)
300     ON DELETE CASCADE;
301 GO

```

Chương 6

Extensions

Phần III

Conclusion

Chương 7

Summary

Chương 8

Future research

Phần IV

Technical appendix

Phụ lục A

Generic technical topic

Phụ lục B

Another technical topic

Phụ lục C

Last technical topic

This is a last appendix with a bit of text.

C.1 A first section

Sed vestibulum ex a tristique lacinia. Integer interdum magna vel magna rutrum fermentum. Maecenas sed mi in nunc convallis rutrum. Vivamus dapibus bibendum est, ac tincidunt ipsum fermentum id. Nunc id est turpis. Suspendisse potenti. Sed ac laoreet nulla, eu sollicitudin libero. Suspendisse tempus orci nec mauris volutpat eleifend. Integer nec tristique libero. Sed vehicula ipsum sit amet magna accumsan, nec pulvinar nisl consequat. In ac hendrerit turpis. Sed aliquet luctus mauris, vitae congue turpis. Nullam blandit lacus vel interdum eleifend.

C.2 A second section

Nam pretium mauris eros, nec sollicitudin risus tincidunt a. Vivamus in augue vitae ligula scelerisque dapibus. Integer eget metus aliquet, efficitur nibh eget, pharetra eros. Suspendisse luctus interdum ex id suscipit. Donec quis augue mauris. Nulla ut erat eget nisl hendrerit malesuada. Vivamus eu tortor sit amet sem fringilla eleifend. In hac habitasse platea dictumst. Suspendisse potenti. Ut eget libero sed orci tempor ullamcorper:

$$S^*(z) = \frac{\alpha}{1 - \gamma(1 - \alpha)} > \int \frac{\alpha(i)}{\gamma(i)} di. \quad (\text{C.1})$$

Nam pretium mauris eros, nec sollicitudin risus tincidunt a. Vivamus in augue vitae ligula scelerisque dapibus. Integer eget metus aliquet, efficitur

nibh eget, pharetra eros. Suspendisse luctus interdum ex id suscipit. Donec quis augue mauris. Nulla ut erat eget nisl hendrerit malesuada. Vivamus eu tortor sit amet sem fringilla eleifend. In hac habitasse platea dictumst. Suspendisse potenti. Ut eget libero sed orci tempor ullamcorper:

$$\mathcal{W}^2(\zeta) = \left[\frac{\beta}{1 - \nu(1 - \xi)} \right] \pm \Lambda \quad (\text{C.2})$$

$$\mathcal{L}^2(\zeta) = \left(\frac{\beta}{1 - \nu(1 - \xi)} \right) \quad (\text{C.3})$$

$$\overline{a - b} = \int_0^{+\infty} \frac{\beta}{1 - \nu(x)(1 - \xi)} dx \in \mathbb{R}. \quad (\text{C.4})$$

C.3 A third section

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed aliquam magna vel urna ultrices, sed lacinia nulla mattis. Fusce non nunc nec est mollis malesuada. Nullam dignissim nulla sit amet libero facilisis, eget fringilla libero sagittis. Suspendisse potenti. Vivamus fermentum consectetur ante, at rhoncus nisi tristique vel. Vivamus in est quis justo fermentum lacinia ac eu leo. Sed ac pulvinar nulla. Etiam quis felis dapibus, vulputate metus eu, finibus nunc. Sed vel sodales dui. Nam venenatis dolor non orci tempus fermentum. Vivamus sodales justo a ligula cursus aliquet. Sed fringilla nunc vitae justo finibus, id placerat lectus sodales.

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Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum, quia dolor sit amet consectetur adipiscing velit, sed quia non numquam eius modi tempora incidunt, ut labore et dolore magnam aliquam quaerat voluptatem. At vero eos et accusamus et iusto odio dignissimos ducimus, qui blanditiis praesentium voluptatum deleniti atque corrupti, quos dolores et quas molestias excepturi sint, obcaecati cupiditate non provident, similique sunt in culpa, qui officia deserunt mollitia animi, id est laborum et dolorum fuga.